

Problem

A four-bit DAC covers a voltage range of 0 to 10 V. Calculate the resolution of the DAC in volts per discrete binary step.

Step-by-step solution

Step 1 of 2

Consider the input and output values of the four bit Digital to Analog converter shown in Table 1.

Table 1

Binary input	Decimal value	Output (Volts)
0000	0	0
0001	1	x
0010	2	$2x$
0011	3	$3x$
0100	4	$4x$
0101	5	$5x$
0110	6	$6x$
0111	7	$7x$
1000	8	$8x$
1001	9	$9x$
1010	10	$10x$
1011	11	$11x$
1100	12	$12x$
1101	13	$13x$
1110	14	$14x$
1111	15	$15x$

Step 2 of 2

Consider that the range of voltage is from 0 V to 10 V.

$$15x = 10$$

Therefore,
$$x = \frac{10}{15}$$

$$= 0.667 \text{ V}$$

Therefore, the resolution of the Digital to Analog converter in volts per discrete binary step is

$$\boxed{0.667 \text{ V}}$$